Do CV-parser errors reach the selection? A fairness audit of the Adecco reverse matching

Francesco Baiocchi
francesco.baiocchi2@studio.unibo.it

Leonardo Petrilli
leonardo.petrilli@studio.unibo.it

Abstract

Recruiting agencies use a CV parser to turn a CV into a structured profile, and a matcher to score the profile against the job offers. In a previous project we found that the proprietary parser used by Adecco misses much of the information written in the CVs. This project asks whether those errors matter: do they change who is selected for a job, and do they penalize some groups (gender, macro-region) more than others?

We work on Adecco's reverse matching, 51,490 (candidate, job) pairs. We first reconstruct the matcher from its outputs: the score is a weighted sum of seven components, and our formula reproduces 99.9% of the scores exactly. We then score every pair twice, with the profile as read by the parser (O) and with the profile corrected with the items we recovered from the CV text (C). We define the selection inside each job's list (top half, top 3) and compare the two selections with the three group fairness criteria.

The parser's errors reach the selection: correcting them changes 8.3% of the outcomes. Independence holds for every group, and there is no gender gap. Separation and sufficiency fail for the South: the decisions on southern candidates are wrong more often, in both directions. A robustness analysis shows that the cause is mostly in the lists, not in the parser: southern jobs more often require no skill that separates the candidates, so the scores pile up near the cut and any noise flips more decisions. The mitigation techniques have little to act on in the real system: equalized odds post-processing could only level the North down to the South. We therefore show pre-, in- and post-processing on a toy hiring case with a known truth.

1 Introduction

1.1 Context and motivation

At Adecco a *parser* reads each CV and extracts a structured profile: driving licenses, languages, past job titles, skills. A *matcher* then compares the profile with the requirements of each job offer and gives a score to every (candidate, job) pair. The recruiter sees the candidates of a job ordered by score.

$$\text{CV} \xrightarrow{\text{parser}} \text{profile} \xrightarrow{\text{matcher}} \text{score of every (candidate, job) pair} \longrightarrow \text{position of the candidate in the job's list}$$

Recruitment is a high-risk use of AI under the EU AI Act (Annex III, point 4(a)) [1]: an error here decides who gets access to a job. If the parser misses a driving license that is written in the CV, the candidate can lose a place in the list of a job that requires it, and nobody notices.

In a previous project we studied the parser alone. We had no access to the parser, only to its input (the anonymized CV text) and its output (the extracted skills). We wrote our own extractors: regular expressions for driving licenses and languages, exact and semantic matching against the ESCO occupation list [2] for job titles. With them we found information that is in the CV text but missing from the parser output, the parser's *false negatives*. Table 1 shows that the parser misses a large part of what our extractors find. The miss rates were similar across gender and macro-region.

Table 1. The parser against our extractors. The driving license is counted per candidate, languages and job titles per item.

field	found by both	missed by the parser	recall of the parser
Driving license	1,689	2,445	0.41
Languages	10,389	5,438	0.66
Job titles	7,838	6,394	0.55

This project follows the errors one step further, down to the selection of the candidates.

1.2 Research questions

- **RQ1.** Do the errors of the CV parser propagate to the selection on the job offers, and do they penalize some groups (gender, macro-region) more than others?
- **RQ2.** If some groups are penalized, which mitigations can reduce the gap, and at what cost?
- **RQ3.** When does a parser error change a selection, and why more often for some groups? How robust are the conclusions to our choices?

1.3 The study in four phases

1. **Data and audit.** We reconstruct how Adecco computes the score, so that we can compute it again with a different profile (Section 3).
2. **Protocol.** We fix what "selected" means, before computing any fairness number (Section 4).
3. **Propagation.** We compare the selection made with the parser's profile and with the corrected one, group by group, and we explain the gap we find (Sections 5 and 6).
4. **Mitigation.** We check what a mitigation could do on the real system, and we show the use of the mitigation techniques on a toy case (Section 7).

2 Data and system under study

2.1 The data

The data come from Adecco and are anonymized. They are available in the GitLab repository of the project. We use:

- the **CVs**, as raw text, and the **parser output**, a list of extracted items per candidate with their type (driving license, language, job title, and others);
- `ReverseMatching.xlsx`, with 51,490 pairs for 7,769 candidates and 9,310 jobs, in three sheets. `ReverseMatching` has one row per (candidate, job) pair, with the `match_score`

and seven components of the score. `Candidates` has the profile that Adecco used for each candidate. `Jobs` has the requirements of each job.

The cleaning of the CVs done in the previous project kept 7,521 candidates with a usable CV, and we keep their pairs. Twenty-two pairs are exact duplicates and are removed.

2.2 A reverse matching read as a direct matching

The *reverse matching* starts from the candidate: it recommends a list of jobs to each candidate. The recruiter uses the opposite view, the *direct matching*: a job and its candidates, ordered by score. Adecco also provides a direct matching file, but it covers other candidates and other jobs and cannot be joined with our CVs. So we read the reverse matching from the job's side: the list of a job is made of the candidates who have that job among their recommendations, ranked by score.

Lists are short. The median job has 3 candidates, and 3,431 of the 9,238 jobs have at least 5.

2.3 Sensitive attributes

The sensitive attributes are **Gender**, **Location** (macro-region: North, Center, South) and their intersection. Table 2 gives the candidates per group. The candidates with gender `Other` (106) or `Unknown` (11) are too few for group statistics and are left out of the fairness tables.

Table 2. Candidates with a clean CV, by gender and macro-region.

	North	Center	South
Female	2,458	505	457
Male	2,885	496	603

Age cannot be used. In the Adecco file `Age_bucket` is a function of `Gender`: all women are 55–74 and all men are 25–34. The column is corrupted, and we never use it.

Each group is compared with a reference group, which plays the role of the historically privileged group in the metrics: `Male`, `NORTH` and `Male_NORTH` (Table 2).

3 Phase 1: reconstructing the matcher

To score every pair again with a different profile, we need to know how the matcher turns a profile into a score. The data give the score and its seven components for every pair, but not the rules that compute them, so we reconstruct these rules from the data.

3.1 The score is a formula

Every component lies in $[0, 1]$. Four are binary (`Interest`, `Card_match`, `Languages_match`, `Regulated_Trainings_match`), `Distance_match` takes three values, `match_skills` and `Experience` are continuous. The score, instead, ranges from 2.0 to 33.8.

Outliers. Most scores are low, and a small group forms a long right tail (Figure 1). With the box-plot rule, a pair is an outlier when its score is above the upper whisker, 6.41. There are 1,110 outliers (2.2% of the pairs), in 143 of the 9,310 jobs.

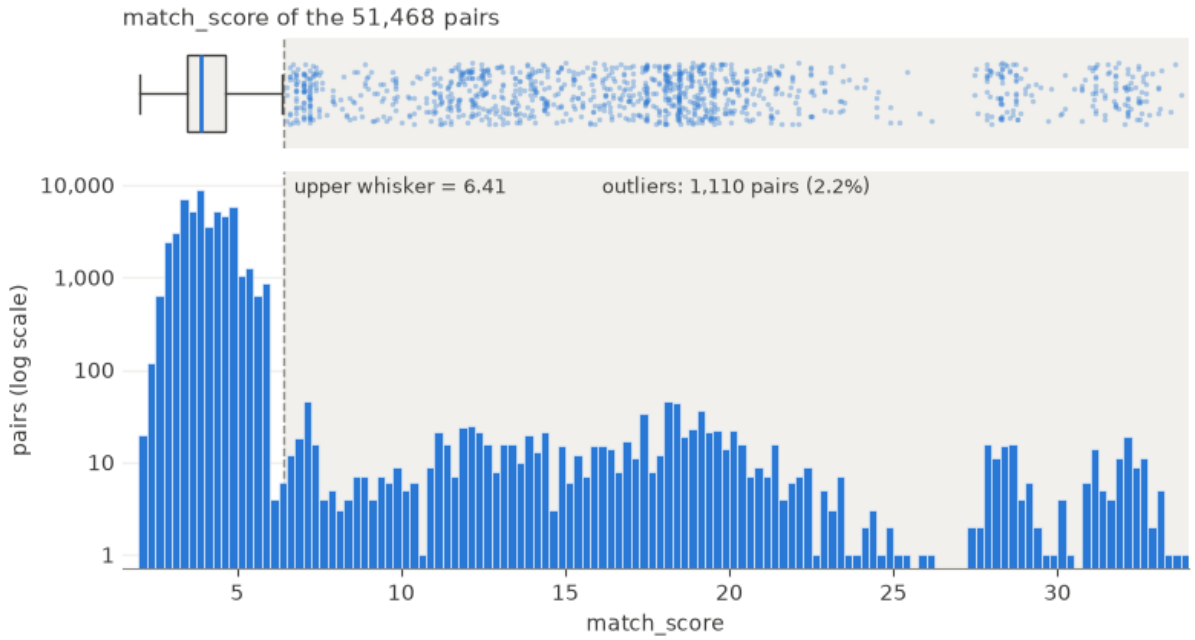


Figure 1. Distribution of `match_score` *(log scale). The dashed line is the upper whisker of the box plot.*

Weights. We fit a linear regression of the score on the seven components, on the pairs without outliers. The coefficients are almost round numbers (Table 3). Rounded to one decimal, they give

$$\text{score} = 2\,\text{Experience} + \text{Interest} + \text{match_skills} + \text{Distance} + 0.3\,(\text{Card} + \text{Languages} + \text{Regulated}).$$

Table 3. Linear regression coefficients of the seven components, and the rounded weights of the formula.

component	estimated, without outliers	rounded	estimated, all pairs
`Experience`	2.04	2	6.08
`Interest`	1.00	1	0.88
`Card_match`	0.30	0.3	0.54
`Languages_match`	0.29	0.3	-0.09
`Regulated_Trainings_match`	0.30	0.3	0.57
`match_skills`	1.00	1	1.12
`Distance_match`	1.00	1	0.49
intercept	-0.02	0	-3.31

Exact check. A good fit is not enough: an approximate formula would leave a small error on every pair. So we count the scores that the rounded formula reproduces exactly (error below 10^{-4}): 99.88% of the 50,358 pairs without outliers. On these pairs the score is the formula, not an approximation of it.

The outliers cannot follow it. With every component at 1 the formula gives $2 + 1 + 1 + 1 + 3 \times 0.3 = 5.9$, below the outlier threshold. Their scores contain something that is not in the seven components and that the data do not show, so we leave those pairs out of the study.

3.2 The rules of the components

Five components can be recomputed from the candidate's profile and the job's requirements (Table 4). We apply our rules to the real profile and compare the result with the value Adecco gives: they agree on more than 99% of the pairs.

Table 4. The rules of five components and their agreement with Adecco's values.

component	value	agreement
`Card_match`	1 if the candidate has at least one of the driving licenses the job lists, or the job lists none; otherwise 0	99.74%
`Languages_match`	the same, with the languages	99.95%
`Regulated_Trainings_match`	the same, with the regulated trainings	99.98%
`match_skills`	the share of the job's skills that the candidate has; 0 if the job lists none	99.92%
`Distance_match`	1 up to 10 km, 0.6 up to 20 km, 0.2 up to 25 km, 0 beyond	100%

The other two components are different:

- `Interest` says whether the job category is among the candidate's categories of interest. These are filled in by hand in the profile, and the parser does not extract them, so `Interest` is the same in every version of the profile.
- `Experience` cannot be recomputed from the data. We estimate it (Section 3.3).

3.3 Estimating `Experience`

Adecco's documentation defines `Experience` as the "highest standardized similarity between the job title and the candidate's experience". Adecco computes it with an internal model that we do not have, so we estimate it by comparing the job category with the candidate's past job titles. The estimate is:

- 1 if the category appears verbatim among the titles;
- 0 if the candidate has no titles;
- otherwise, the highest cosine similarity between the category and a title, computed by a sentence encoder.

Choosing the encoder. On the real profile we know Adecco's `Experience` for every pair, so we can try several encoders and keep the one that reproduces it best (Table 5). The texts are mostly Italian, but not only: the parser's job titles follow an English taxonomy and some were never translated, and the recruiters write part of the job categories in English. In the real profile, 16% of the titles and 5% of the job categories are in English. So we compare several multilingual encoders, and we include an Italian-only encoder in the comparison to measure what it costs to read only Italian. We also add a string-matching baseline, with no model, to check whether an encoder does better than plain string matching.

Table 5. Estimators of `Experience` *on the real profile. Spearman: rank correlation with Adecco's value. MAE: mean absolute error. Lowest estimate: on the pairs whose candidate has titles, where Adecco's lowest value is 0.83. Gross errors: pairs estimated below 0.4 while Adecco gives more than 0.8.*

estimator	parameters (M)	Spearman	MAE	lowest estimate	gross errors
lexical baseline	–	0.806	0.278	0.13	4,625
MiniLM-L12	118	0.765	0.135	0.23	504
mpnet-base	278	0.847	0.106	0.20	189
e5-base	278	**0.897**	**0.017**	0.81	**0**
e5-large	560	0.876	0.021	0.76	0
sentence-BERTino (Italian only)	68	0.896	0.137	0.10	863

`multilingual-e5-base` [3] has the highest rank correlation, the smallest error and no gross error. It is also on Adecco's scale: its lowest estimate is just below Adecco's lowest value (Figure 2). Size does not explain the differences: the largest model is not better, and the smallest multilingual one ranks the pairs worse than the string baseline. The Italian-only encoder ranks the pairs almost as well as `e5-base`, but its error is eight times larger and it fails across languages: it gives 0.39 to `warehouse worker` and `magazziniere`, the same job, where `e5-base` gives 0.87.

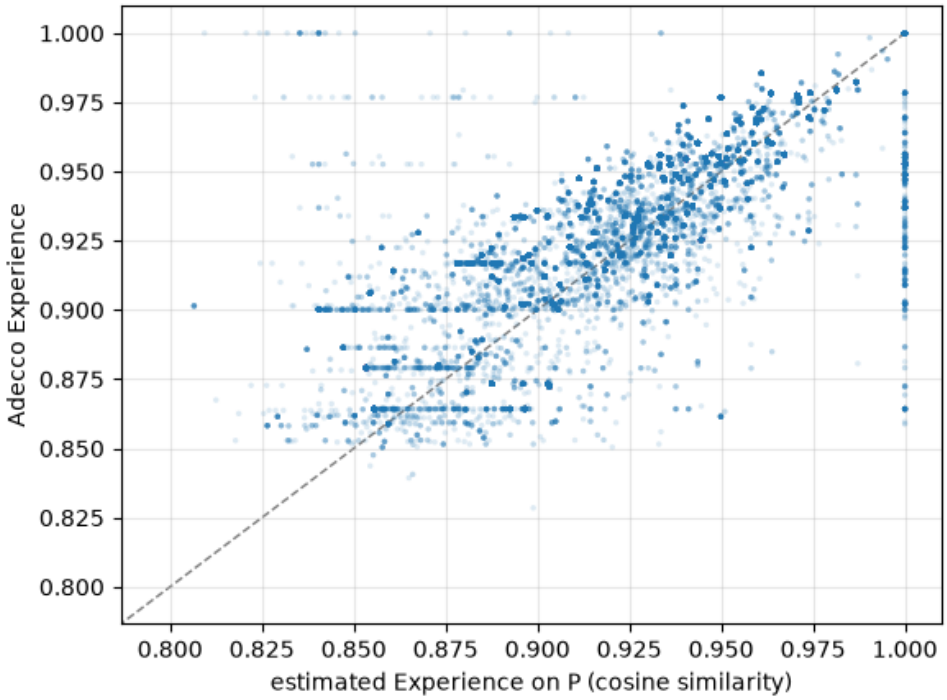


Figure 2. Estimate of `e5-base` *against Adecco's* `Experience`*, one dot per pair of the real profile whose candidate has titles.*

We use the same estimate for every version of the profile: the real profile is also scored with the *estimated* `Experience`, so the three versions are scored in exactly the same way and differ only in the titles.

3.4 Three versions of every profile

Table 6. The three versions of a candidate's profile.

version	content	role
P	the real Adecco profile (`Candidates` sheet)	validates the reconstructed matcher
O (Original)	licenses, languages and job titles only as read by the parser	the fully automatic pipeline
C (Corrected)	O plus the false negatives recovered from the CV text	the reference for the parser's errors

O takes licenses, languages and job titles from the parser output. Skills and regulated trainings are copied from P: the parser does not extract regulated trainings, and names skills differently from the jobs, so they would almost never match. C adds to O the false negatives of the previous project.

The real profile is not the parser output. Table 7 compares P and O candidate by candidate. Job titles come from the parser: every candidate with parsed titles also has titles in P, and half of them share at least one. Licenses and languages do not: only a minority of the candidates with a parsed license have a license in P. So in the real system licenses and languages were entered by hand.

Table 7. P against O, for the candidates where the parser found at least one item of the field. For example, if the parser finds the titles `cassiere` *and* `commesso` *and P has* `commesso` *and* `magazziniere`*, the candidate counts in "also in P" and in "common item", but not in "identical". The last two columns are shares of the candidates counted in "also in P".*

field	candidates with parsed items	also in P	common item	identical
licenses	2,161	17.5%	46.0%	40.5%
languages	6,827	39.3%	17.9%	3.4%
job titles	7,088	100%	50.4%	0.03%

This has two consequences for the design:

- We do not know where the manual fields of P come from, so we do not use P to judge the parser. The bias of the parser is measured only on the comparison between O and C. P is used to validate the rules and the estimate.
- For job titles, O is close to the real system: there too the parser's titles enter the profile, and so the score. For licenses and languages it is not: in the real system they are entered by hand, while O takes them from the parser. Here O shows what would happen if the system relied only on the parser.

Phase 1 ends with a dataset of 48,816 pairs (7,517 candidates, 9,180 jobs), each with its components and its score under P, O and C. From O to C the score changes in 26.3% of the pairs: `Card_match` changes in 10.0%, `Languages_match` in 6.5% and the estimated `Experience` in 11.2% (by more than 0.01).

4 Phase 2: the selection protocol

The data give a score for every pair, but no decision. So, before computing any fairness metric, we define what counts as a selection.

- **Unit.** A (candidate, job) pair: each pair is one decision.

- **Selected.** A candidate is selected for a job if they are in the top half of that job's candidates, ranked by score. A score means something only against the other candidates of the same job: a job with few requirements gives high scores to everyone. So we rank inside each job instead of using one threshold for all jobs. As a stricter second outcome we use the top 3.
- **Ties.** Candidates of the same job with the same score are ordered at random. Each pair gets one random number, used with every score. So every job with n candidates selects exactly $\lceil n/2 \rceil$ of them, and a pair can change outcome between two versions only because the scores change, since the draw stays the same.
- **Lists.** We keep the jobs with at least 5 candidates. A position in a list of one or two candidates says nothing about selection.
- **Scores.** Adecco's own score, and our formula on P, O and C.

The protocol keeps 36,760 pairs (75.3%), in 3,344 jobs, for 6,510 candidates. The median job has 9 candidates. The North holds most of the data (Table 8).

Table 8. Size of the groups in the protocol.

group	pairs	candidates
Female, North	13,798	2,290
Male, North	15,346	2,580
Female, Center	1,669	402
Male, Center	1,631	360
Female, South	2,022	397
Male, South	2,294	481
all	36,760	6,510

Almost every list has candidates from one region only: 3,158 of the 3,344 lists (94.4%), with 93.7% of the pairs. So a southern candidate competes almost only with other southern candidates. Gender is different: women and men are in the same lists, and only 15.8% of the lists have one gender.

Table 9 looks at the outcomes under the four scores.

Table 9. Per score: the share of pairs whose outcome is decided by the random draw (tied with another candidate across the cut), and the share with the same outcome as with Adecco's score.

score	decided by the draw, top half	same outcome as Adecco, top half	decided by the draw, top 3	same outcome as Adecco, top 3
Adecco	20.8%	100%	17.4%	100%
P	18.2%	96.9%	15.4%	97.5%
O	6.0%	81.5%	4.1%	86.1%
C	5.7%	84.2%	3.9%	87.9%

P agrees with Adecco on about 97% of the pairs, with both outcomes, so the formula, the rules and the estimate together reproduce Adecco's decisions. Many of Adecco's decisions are a tie: with its score, one pair in five is tied with another candidate exactly at the cut.

Going from O to C, that is, correcting the parser, changes the outcome (top half) of 8.3% of the pairs. A job with n candidates always has $\lceil n/2 \rceil$ places, whatever the score, so a candidate who enters the top half pushes another one out: 1,522 pairs enter and 1,522 leave. For the same reason 52.4% of the pairs are selected with every score. The correction changes who is selected, not how many.

5 Phase 3: propagation and group fairness

This section answers RQ1. We first look at what the correction changes (Section 5.2). Then we compare the groups with the three criteria of group fairness: independence (Section 5.3), separation (Section 5.4) and sufficiency (Section 5.5). Section 5.6 repeats the comparison with the top 3.

5.1 Method

Decision and label. The decision D is the outcome with the parser's profile O. The label Y is the outcome with the corrected profile C. C is not the truth: it holds what our extractors recovered from the CV text. So the errors below are errors of O relative to C.

Intervals. A group has a few hundred to a few thousand candidates, so its numbers are uncertain. We measure this with a bootstrap: 1,000 times, we draw the candidates again with replacement, draw the ties again and recompute every number. The 95% interval holds the middle 95% of the 1,000 values. We draw candidates, not pairs: the pairs of a candidate share the same profile and group, so they are not independent, and drawing them one by one would make the intervals too narrow. When the interval of a gap contains 0 (or 1 for a ratio), the gap is not statistically significant: the data are not enough to say that the group differs from its reference group.

In the tables, gaps are in percentage points and the interval is in brackets.

5.2 What the correction changes

The correction adds at least one job title to 47.1% of the protocol candidates, a language to 41.1% and a license to 33.4%. A license or a language counts only when the job requires it, so the two matches change in less than one pair in ten (Table 10). The score changes in about one pair in four, in every group: the correction touches the South as often as the North.

Table 10. Share of the pairs of each group where the correction changes a component or the score (%).

group	pairs	`Card_match`	`Languages_match`	`Experience`	score
Female	17,489	9.4	6.9	10.3	25.7
Male	19,271	9.9	6.5	13.0	27.9
North	29,144	10.0	6.6	11.3	26.8
Center	3,300	8.9	7.8	12.4	27.6
South	4,316	8.0	6.5	14.0	26.6

The outcome changes for 3,044 pairs (8.3%). Table 11 says why.

Table 11. Among the pairs whose outcome changes: the share where each component changes (a pair can have more than one), and the share whose own score does not change (%).

group	changed pairs	`Card_match`	`Languages_match`	`Experience`	score unchanged
Female	1,371	19.8	11.5	32.5	45.2
Male	1,673	17.8	9.4	39.3	43.5
North	2,300	19.3	10.4	35.7	44.3
Center	287	13.6	14.3	40.8	42.2
South	457	18.4	7.4	35.7	45.5

In every group, the component that changes most often is `Experience`: the correction adds a job title, and the new title changes the estimate. It is also the component that counts most,

with weight 2 in the formula. Job titles are the only field of the parser that enters the real profile (Section 3.4), so these are the parser's errors that matter in the real system.

A pair can also change outcome because of the other candidates of its list. 1,347 of the 3,044 pairs (44%) keep the same score. The correction only adds information, so no score goes down, and all of these pairs leave the top half: they are candidates overtaken by a competitor whose CV was corrected. With O they had a place only because the parser had missed something in someone else's CV.

5.3 Independence

Independence (statistical parity) asks that every group is selected at the same rate. We compare each group with its reference group in two ways: the statistical parity difference (SPD) and the disparate impact (DI). Figure 3 shows the selection rate of every group with Adecco's score, with O and with C, and Table 12 gives the SPD and the DI.

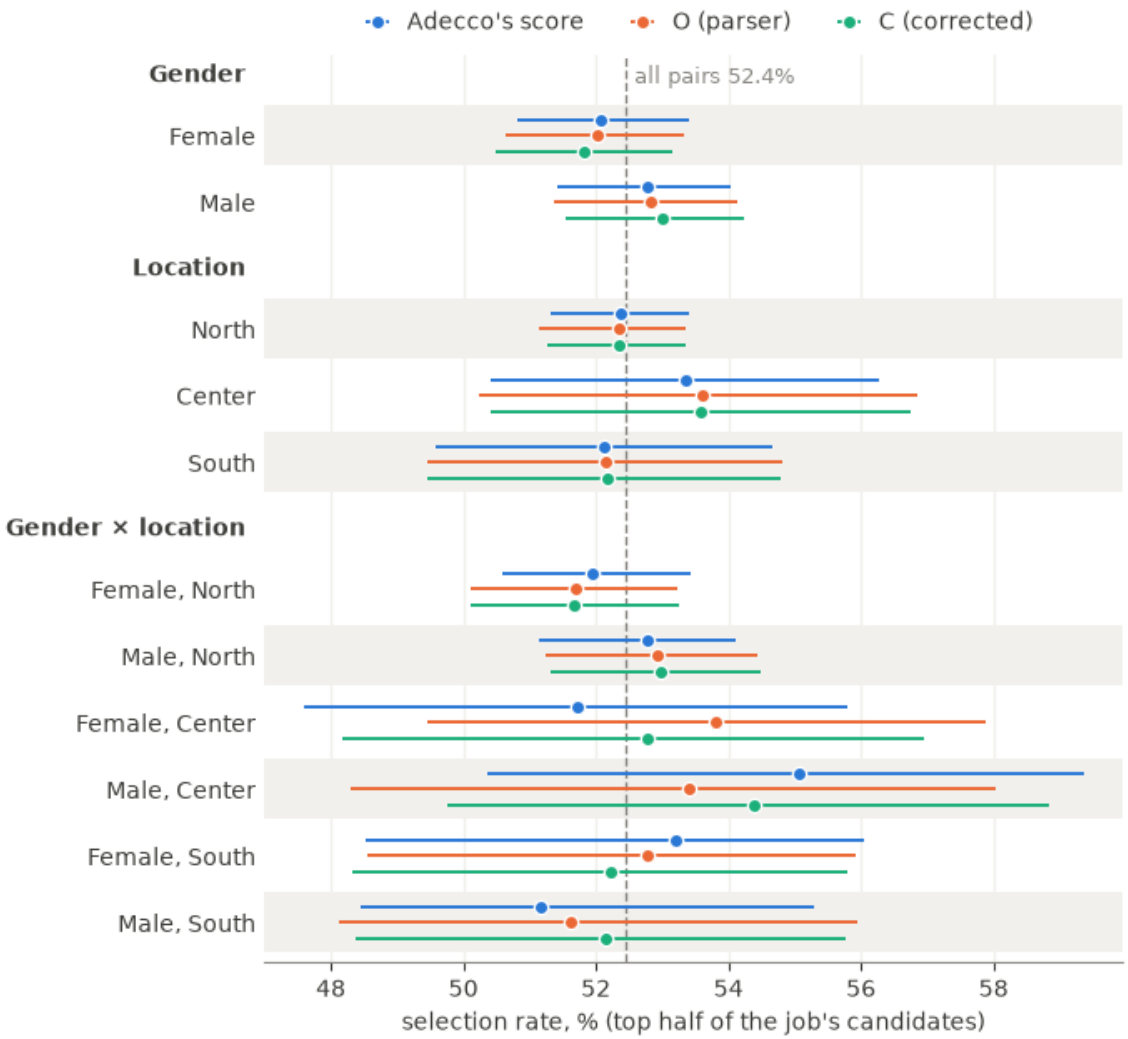


Figure 3. Selection rate (top half) per group, with 95% intervals. The dashed line is the rate of all pairs.

Table 12. Independence: SPD (percentage points) and DI of each group against its reference group (`Male`, `NORTH`, `Male_NORTH`), with the three scores.

group	Adecco's score	O (parser)	C (corrected)
SPD			
Female	-0.7 [-2.6, 1.2]	-0.8 [-2.9, 1.1]	-1.2 [-3.1, 0.8]
Center	1.0 [-2.0, 4.2]	1.3 [-2.3, 4.6]	1.2 [-2.1, 4.5]
South	-0.2 [-2.9, 2.5]	-0.2 [-3.1, 2.8]	-0.2 [-3.0, 2.7]
Female, North	-0.8 [-2.7, 1.4]	-1.2 [-3.4, 1.1]	-1.3 [-3.6, 0.9]
Female, Center	-1.1 [-5.2, 3.5]	0.9 [-3.8, 5.5]	-0.2 [-4.9, 4.3]
Male, Center	2.3 [-2.6, 6.9]	0.5 [-4.6, 5.4]	1.4 [-3.3, 6.0]
Female, South	0.4 [-4.1, 3.5]	-0.2 [-4.7, 3.3]	-0.7 [-4.9, 3.1]
Male, South	-1.6 [-4.5, 3.0]	-1.3 [-5.4, 3.6]	-0.8 [-4.9, 3.4]
DI			
Female	0.99 [0.95, 1.02]	0.98 [0.95, 1.02]	0.98 [0.94, 1.02]
Center	1.02 [0.96, 1.08]	1.02 [0.96, 1.09]	1.02 [0.96, 1.09]
South	1.00 [0.94, 1.05]	1.00 [0.94, 1.05]	1.00 [0.94, 1.05]
Female, North	0.98 [0.95, 1.03]	0.98 [0.94, 1.02]	0.98 [0.93, 1.02]
Female, Center	0.98 [0.90, 1.07]	1.02 [0.93, 1.11]	1.00 [0.91, 1.08]
Male, Center	1.04 [0.95, 1.12]	1.01 [0.91, 1.10]	1.03 [0.94, 1.12]
Female, South	1.01 [0.92, 1.07]	1.00 [0.91, 1.06]	0.99 [0.91, 1.06]
Male, South	0.97 [0.92, 1.06]	0.97 [0.90, 1.07]	0.98 [0.91, 1.06]

Every SPD interval contains 0 and every DI interval contains 1. Women are selected a little less often than men with every score, but well inside the interval. The lowest bound of any DI is 0.90, far above the 0.8 of the four-fifths rule. The data show no violation of independence, before or after the correction.

For the regions this is expected. Almost every list has candidates from one region only (Section 4), and every list selects half of its candidates, so each region is selected at about the same rate whatever the score. For gender it is a result: women and men are in the same lists, and a score could select one more often than the other.

5.4 Separation

Separation (equalized odds) asks that the decision makes the same errors in every group. With C as the label, the false negative rate (FNR) is the share of the pairs selected with C that O rejects, and the false positive rate (FPR) is the share of the pairs not selected with C that O selects. Table 13 gives the two rates of every group and their gaps from the reference group, and Figure 4 shows the gaps.

Table 13. Separation: false negative and false positive rates of O against C (%), and their gaps from the reference group (percentage points).

group	FNR	FPR	FNR gap	FPR gap
Female	7.4 [6.4, 8.4]	8.3 [7.5, 9.0]	-1.0 [-2.4, 0.6]	-0.7 [-1.9, 0.3]
Male	8.4 [7.2, 9.4]	9.1 [8.2, 9.9]	reference	reference
North	7.5 [6.7, 8.3]	8.3 [7.6, 8.8]	reference	reference
Center	8.1 [5.7, 10.9]	9.4 [7.4, 11.8]	0.6 [-2.0, 3.4]	1.1 [-1.0, 3.8]
South	10.2 [8.0, 12.7]	11.0 [9.1, 13.0]	**2.6 [0.4, 5.3]**	**2.8 [0.8, 4.9]**
Female, North	7.2 [6.1, 8.3]	7.8 [6.9, 8.6]	-0.6 [-2.2, 1.0]	-0.9 [-2.0, 0.3]
Male, North	7.8 [6.6, 9.0]	8.7 [7.8, 9.5]	reference	reference
Female, Center	5.2 [2.5, 8.7]	8.0 [5.2, 11.4]	-2.6 [-5.7, 0.9]	-0.7 [-3.7, 3.1]
Male, Center	10.9 [6.9, 15.3]	10.9 [7.5, 14.8]	3.1 [-0.9, 7.8]	2.2 [-1.3, 6.4]
Female, South	10.2 [7.5, 13.7]	12.3 [8.9, 14.6]	2.4 [-0.7, 6.2]	**3.6 [0.1, 6.2]**
Male, South	10.1 [6.8, 13.6]	9.9 [7.9, 13.6]	2.3 [-1.2, 6.0]	1.2 [-0.8, 5.1]

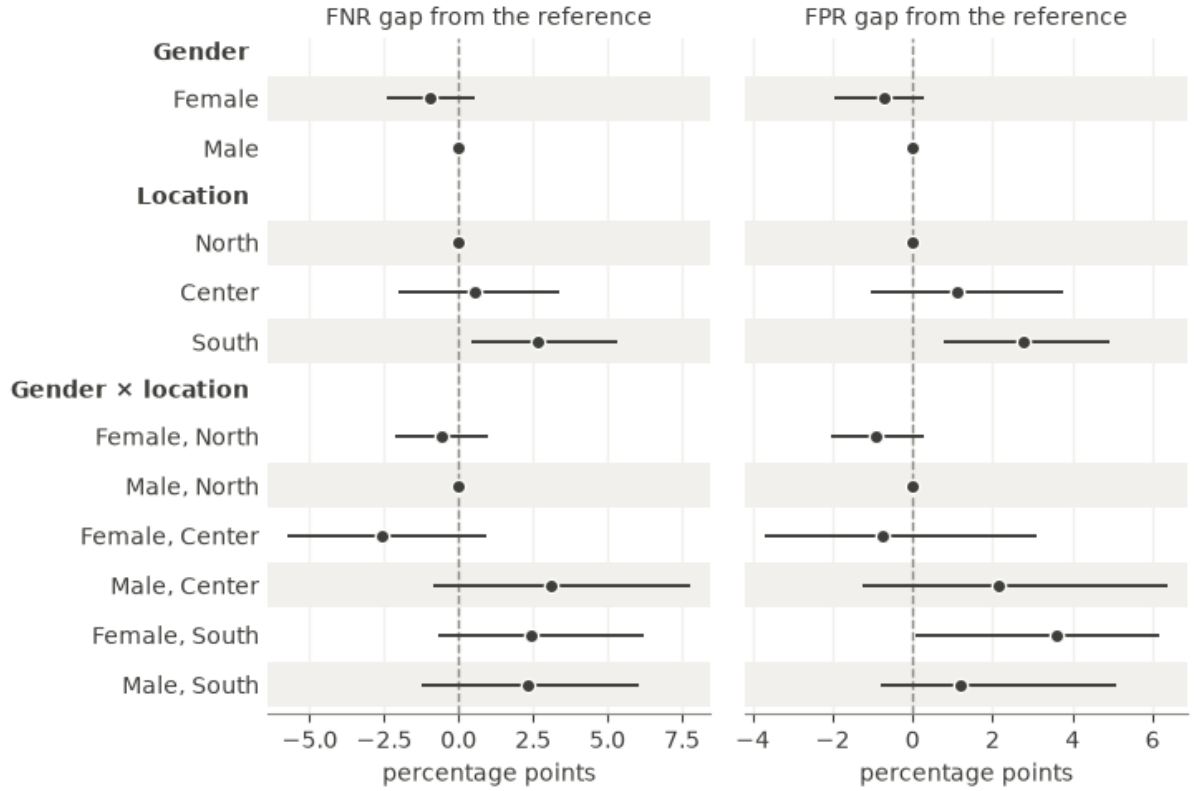


Figure 4. FNR and FPR gaps from the reference group, with 95% intervals.

The South is the only area whose errors differ from the North, in both directions. O leaves out more of the southern pairs that C selects (FNR), and selects more of the southern pairs that C does not select (FPR). Each gap is about three points, and both intervals exclude 0. Among the intersectional groups, only the FPR of women from the South excludes 0. Between women and men there is no gap.

5.5 Sufficiency

Sufficiency (calibration) asks that a decision is equally reliable in every group. The positive predictive value (PPV) is the share of the pairs selected by O that C also selects, and the negative predictive value (NPV) is the share of the pairs rejected by O that C also rejects.

Table 14 gives the two values of every group and their gaps from the reference group, and Figure 5 shows the gaps.

Table 14. Sufficiency: predictive values of O against C (%), and their gaps from the reference group (percentage points).

group	PPV	NPV	PPV gap	NPV gap
Female	92.3 [91.6, 93.1]	92.0 [91.0, 93.1]	0.3 [-0.7, 1.5]	1.4 [-0.1, 3.0]
Male	91.9 [91.2, 92.7]	90.6 [89.4, 91.8]	reference	reference
North	92.5 [92.0, 93.1]	91.7 [90.9, 92.6]	reference	reference
Center	91.9 [89.7, 93.5]	90.7 [87.6, 93.4]	-0.6 [-2.9, 1.1]	-1.1 [-4.3, 1.9]
South	89.9 [88.1, 91.6]	88.9 [86.3, 91.2]	**-2.6 [-4.4, -0.9]**	**-2.8 [-5.6, -0.2]**
Female, North	92.7 [91.9, 93.5]	92.3 [91.1, 93.4]	0.5 [-0.7, 1.5]	1.1 [-0.6, 2.8]
Male, North	92.2 [91.5, 93.1]	91.2 [89.8, 92.6]	reference	reference
Female, Center	93.0 [90.0, 95.4]	94.0 [90.1, 97.0]	0.7 [-2.3, 3.3]	2.8 [-1.4, 6.3]
Male, Center	90.7 [87.5, 93.5]	87.2 [82.1, 91.6]	-1.5 [-4.9, 1.3]	-4.0 [-9.4, 0.5]
Female, South	88.8 [86.8, 91.7]	88.7 [85.0, 91.8]	**-3.4 [-5.5, -0.5]**	-2.5 [-6.5, 0.9]
Male, South	90.8 [87.8, 92.6]	89.1 [85.7, 92.5]	-1.5 [-4.8, 0.4]	-2.1 [-5.9, 1.5]

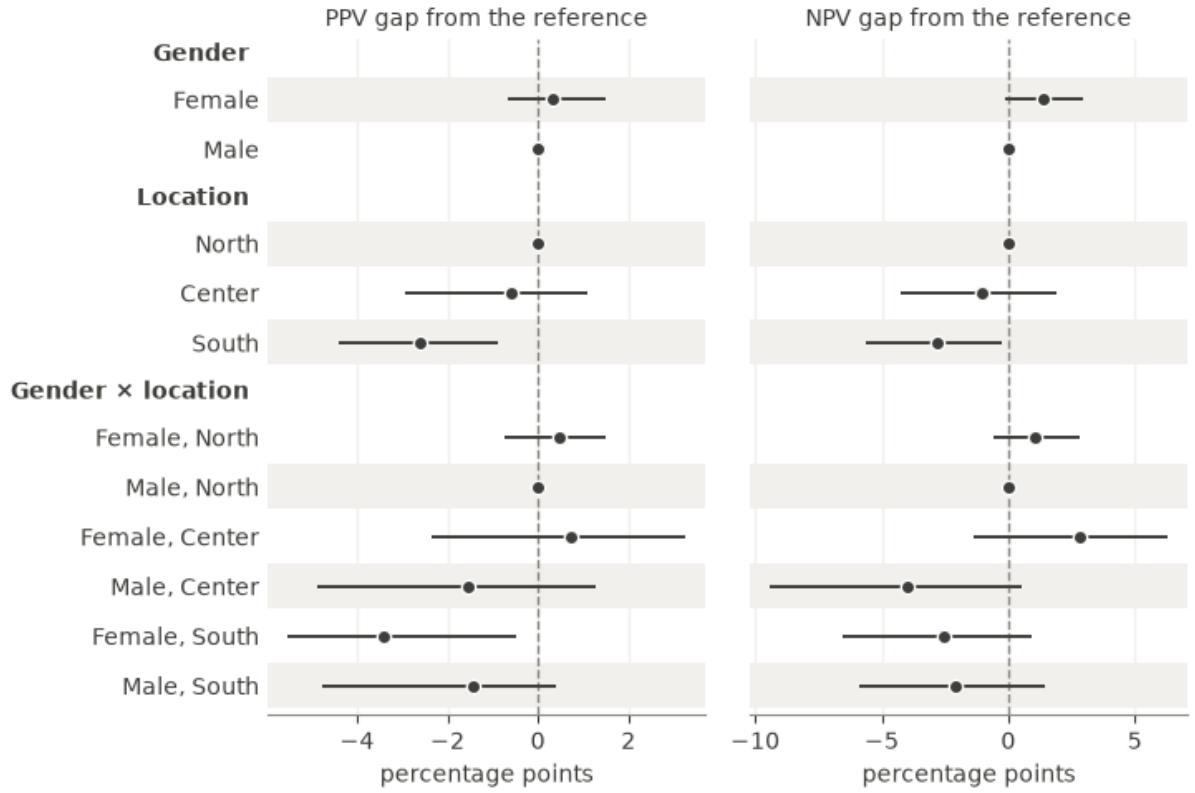


Figure 5. PPV and NPV gaps from the reference group, with 95% intervals.

Both predictive values are high in every group: when the parser's profile selects or rejects a candidate, the corrected one confirms it about nine times in ten. The South is again the only area that differs from the North, on both sides: its selections and its rejections are confirmed a few points less often. Women from the South have a lower PPV. Between women and men there is no gap.

For the regions, sufficiency gives the same result as separation: O and C select the same number of pairs in every list, and almost every list has one region (Section 4), so the PPV and NPV gaps are the FNR and FPR gaps with the opposite sign.

So the South passes independence but fails separation and sufficiency: it is selected at the same rate, but the decisions on southern candidates are wrong more often.

5.6 The same with the top 3

A stricter selection: the first three candidates of every job. Table 15 repeats the gaps of Tables 13 and 14 with the top 3 as the outcome.

Table 15. The gaps of Tables 13 and 14 with the top 3 as the outcome (percentage points).

group	FNR gap	FPR gap	PPV gap	NPV gap
Female	-0.4 [-2.6, 2.2]	**-0.9 [-1.6, -0.1]**	**2.4 [0.6, 4.2]**	0.0 [-0.9, 0.9]
Center	1.8 [-2.3, 6.6]	**1.9 [0.4, 3.9]**	-1.8 [-5.6, 1.4]	-1.9 [-4.3, 0.1]
South	**4.4 [0.4, 8.3]**	0.5 [-0.4, 1.6]	**-4.5 [-7.9, -1.9]**	-0.4 [-1.8, 0.9]
Female, North	0.5 [-2.0, 3.1]	-0.7 [-1.4, 0.1]	1.9 [-0.2, 3.7]	-0.3 [-1.3, 0.7]
Female, Center	-2.2 [-6.6, 3.6]	0.0 [-1.8, 1.9]	2.6 [-1.2, 6.4]	-0.1 [-3.1, 2.0]
Male, Center	6.4 [-0.3, 13.9]	**3.3 [0.8, 6.8]**	-4.6 [-11.4, 0.1]	**-4.0 [-7.8, -1.0]**
Female, South	5.5 [-0.1, 11.7]	-0.2 [-1.5, 1.4]	-2.7 [-7.8, 1.6]	-0.8 [-3.0, 1.0]
Male, South	3.9 [-1.2, 9.4]	0.4 [-1.0, 2.2]	-4.4 [-9.3, 0.1]	-0.3 [-2.3, 1.3]

With the top 3, two of the South's four gaps remain: the higher FNR, larger than with the top half, and the lower PPV. Both concern the selected pairs: O leaves out more of the southern pairs that C selects, and C confirms fewer of the southern pairs that O selects.

The South's FPR and NPV gaps do not remain. Both are measured on the pairs that are not selected. With three places per job these are most of the pairs, and O and C agree on almost all of them, in the South as in the North. So the two regions differ only on the selected pairs.

A few other gaps exclude 0 with one outcome only: for example a higher FPR in the Center here, and the lower PPV of women from the South in Table 14.

Tables 13 to 15 look at 64 gaps (8 groups, 4 rates, 2 outcomes). With a 95% interval, one gap in twenty excludes 0 by chance alone, even when there is no real difference: about three of the 64. So we do not trust a gap that appears with one outcome only. Two gaps appear with both outcomes (top half, top 3): the higher FNR and the lower PPV of the South, the same gap seen by separation and by sufficiency.

6 Why the South: the margins and the job lists

The correction touches southern and northern pairs equally often (Table 10), but it flips more southern outcomes. This section explains why, and answers the first part of RQ3: when a parser error changes a selection, and why more often for some groups.

An outcome is a place in a list. A change of score flips the outcome only if it moves the pair across the cut: a pair far from the cut keeps its place, a pair close to the cut loses it for a small change. This is a question of robustness, that is, of how stable a decision is when its input changes a little.

The section uses one metric, the **error rate**: the share of a group's pairs whose outcome differs between O and C. It is 10.6% in the South and 7.9% in the North, a gap of 2.7 points (Table 17).

6.1 Southern pairs are closer to the cut

The *margin* of a pair is the distance between its score and the score of the last selected candidate of its list.

Table 16. Margins per region, with O: median margin, share of pairs within 0.1 of the cut, and share of pairs that Adecco's own score ties across the cut.

region	median margin	within 0.1	tied across the cut (Adecco)
North	0.35	31.4%	19.5%
Center	0.30	34.8%	21.0%
South	0.16	42.5%	29.1%

The median margin of the South is less than half the North's. Adecco's own score ties about three southern pairs in ten across the cut, against two in ten in the North.

6.2 With the North's margins, half of the gap goes

We split the pairs into five bands of margin and weight the error rate of each region in each band by the share of northern pairs in that band. The result is the error rate the region would have if its pairs were as far from the cut as the North's.

Table 17. Error rate (%), observed and with the North's margins, and gaps from the North (percentage points). Explained is the difference between the two gaps.

region	error	error with the North's margins	gap	gap with the North's margins	explained
North	7.9 [7.3, 8.4]	7.9 [7.3, 8.4]	reference	reference	reference
Center	8.7 [7.1, 10.6]	8.3 [6.7, 10.1]	0.8 [-0.8, 2.8]	0.4 [-1.3, 2.3]	0.4 [0.0, 0.8]
South	10.6 [9.1, 12.1]	9.4 [8.1, 10.8]	2.7 [1.2, 4.3]	1.5 [0.1, 3.1]	1.2 [0.7, 1.7]

Being close to the cut explains about half of the South's extra errors: 1.2 of 2.7 points, with an interval that excludes 0.

6.3 Why the margins are small: jobs with no skill that separates the candidates

A term of the score helps to rank the candidates of a job only if they have different values of it. Table 18 gives, for every term, how much it varies inside a list. Almost every list has candidates from one region only (Section 4), so we can speak of southern and northern lists.

Table 18. Standard deviation of each term of the score among the candidates of the same job, averaged over the pairs of each region (points of score, with O). Change: the South against the North.

term	North	Center	South	change
`Experience`	0.34	0.30	0.39	+13.3%
`Interest`	0.32	0.29	0.25	-20.4%
`Card_match`	0.03	0.03	0.03	0%
`Languages_match`	0.02	0.03	0.02	0%
`Regulated_Trainings_match`	0.00	0.00	0.00	0%
`match_skills`	0.18	0.11	0.07	-61.3%
`Distance_match`	0.24	0.26	0.23	-6.6%

`match_skills` is the term that changes most: inside a southern list it varies less than half as much as inside a northern one. `Interest` varies less too. These are the two terms that the parser never touches, so the parser is not what compresses the southern lists.

`match_skills` is the share of the job's required skills that the candidate has. When the job requires no skill, or skills that give every candidate the same value, the term is the same for

everyone and the scores pile up near the cut. So we split the lists into three classes (Table 19).

Table 19. Lists per class and region, with the median margin.

class of list	North: share of lists	North: median margin	South: share of lists	South: median margin
no skill required	41.9%	0.30	59.9%	0.11
skills required, same for all	13.7%	0.21	15.1%	0.08
skills that separate the candidates	44.3%	0.40	25.0%	0.41

Six southern lists in ten belong to a job that requires no skill, against four in ten in the North. Where the required skills separate the candidates, the margins are the same in the two regions. In the other two classes the South's margins are much smaller.

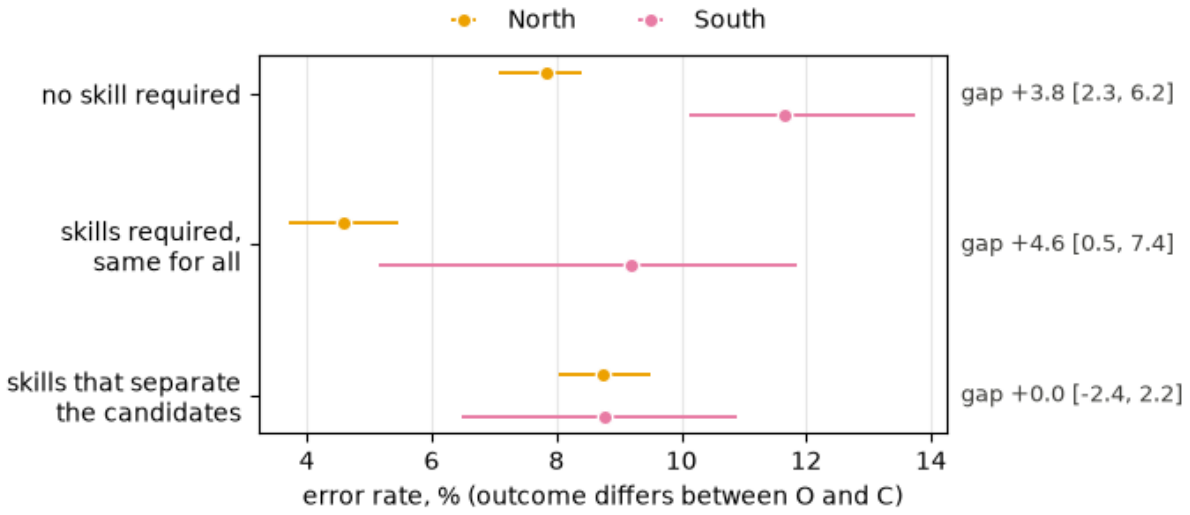


Figure 6. Error rate of the North and of the South in each class of list, with 95% intervals. The gap is on the right.

The gap appears only where the score cannot separate the candidates (Figure 6). In the first two classes the gap is about four points and its interval excludes 0. Where the required skills separate the candidates, the gap is 0.0 points and its interval contains 0: there the data show no difference between the South and the North.

6.4 Random noise flips more southern decisions too

If the gap comes from the lists, any small change of the scores should flip more southern decisions, even with no error of the parser. We test it with random noise. We start from the real profile P, which our correction does not change. We add Gaussian noise to the estimated `Experience` of every pair, drawn in the same way in every region. We compute the scores and the ranking again and count the pairs whose outcome changes. Table 20 gives the mean of 20 repetitions.

Table 20. Share of decisions that flip under random noise on `Experience` (%), for two noise levels σ.

class of list	North, $\sigma = 0.05$	South, $\sigma = 0.05$	North, $\sigma = 0.10$	South, $\sigma = 0.10$
no skill required	12.3	18.8	15.4	23.1
skills required, same for all	13.7	22.7	16.8	26.6
skills that separate the candidates	8.4	11.6	10.6	13.5

Southern decisions flip more often than northern ones in every class. The difference is largest in the first two classes, 6 to 10 points, and about half as large in the third. So errors that are the same in every region are enough to produce a gap: the parser does not need to make more errors on southern CVs.

7 Mitigation

This section answers RQ2: which mitigations can reduce the South's gap, and at what cost? We first look at what the techniques of the course can do on the real system (Section 7.1), then test them on a toy case where they can act (Section 7.2). The toy is only a demonstration, and its numbers say nothing about Adecco's data.

7.1 On the real system

The bias found in RQ1 is a gap in separation and sufficiency. Table 21 compares what each mitigation technique needs with what the real system has.

Table 21. The mitigation techniques and the real system.

technique	what it needs	what the real system has
Pre-processing (reweighing)	a training set with labels	nothing is trained: the weights of the formula are fixed
In-processing (Prejudice Remover)	a loss to add the fairness term to	no loss
Post-processing on the score	a threshold on a score, and true labels	a quota per list, not a threshold
Post-processing on the decisions	decisions and labels	can be tested, with C as the label

So we test the last one. Equalized odds post-processing [4] flips decisions at random: a selected pair stays selected with probability `keep`, a rejected pair becomes selected with probability `add`, both per region. A region with rates $(\mathrm{FPR}, \mathrm{TPR})$, where $\mathrm{TPR} = 1 - \mathrm{FNR}$, ends with

$$\mathrm{TPR}' = \mathrm{keep} \cdot \mathrm{TPR} + \mathrm{add} \cdot (1 - \mathrm{TPR}), \qquad \mathrm{FPR}' = \mathrm{keep} \cdot \mathrm{FPR} + \mathrm{add} \cdot (1 - \mathrm{FPR}).$$

So a region can reach only the rates inside a quadrilateral (Figure 7), and equalized odds needs a point inside the quadrilaterals of all regions.

Table 22. Equalized odds post-processing of the decisions of O, with C as the label (rates in %).

region	TPR	FPR	keep	add	TPR after	FPR after
North	92.5	8.3	0.97	0.03	89.8	11.1
Center	91.9	9.4	0.98	0.02	89.8	11.1
South	89.8	11.0	1.00	0.00	89.8	11.1

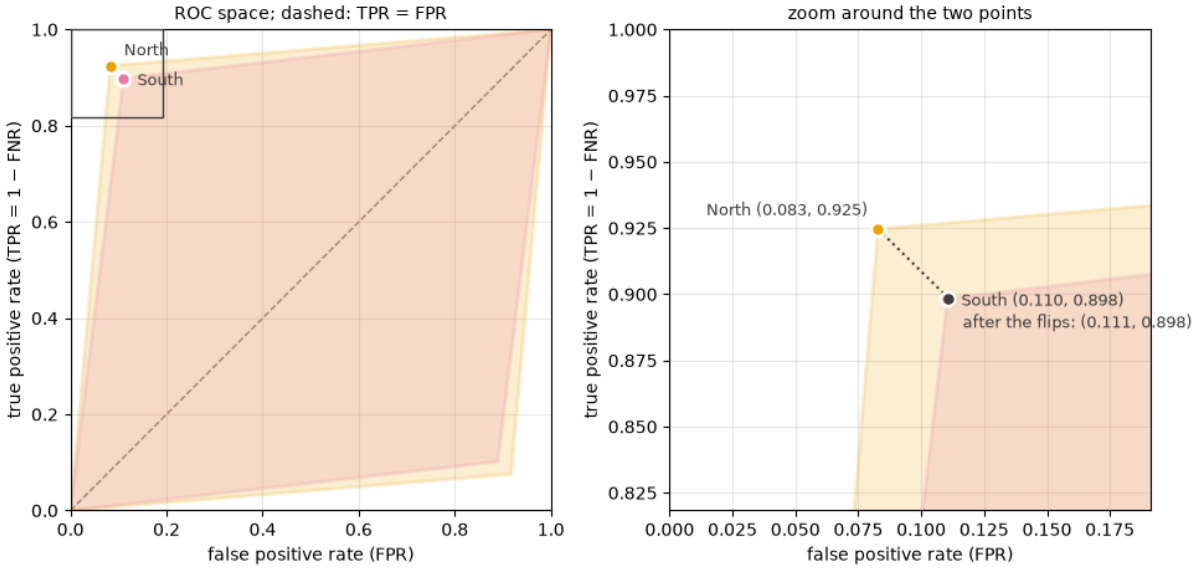


Figure 7. The rates each region can reach by flipping its decisions at random. The dotted line is the move of the North to the common point.

The South's point lies inside the North's quadrilateral, so the best rates the regions can share are the South's own. The optimizer leaves every southern decision as it is and brings the North and the Center down to the South's rates. This is leveling down [5]: the gap closes, and no southern decision gets better.

7.2 On a toy case

The techniques need what the real system lacks (Table 21), so we test them on a toy case where they can act: one job opening in the North, 20,000 candidates for training and 20,000 for evaluation, with a known truth.

- Each candidate has two scores, `skills` and `experience`, independent of the region, and a `distance` from the opening. Southern candidates live farther, so the distance tells the region, with some overlap.
- The **truth** depends only on the two scores, with a little noise.
- The **past decisions** follow the truth with one exception: the recruiters rejected 30% of the qualified southern candidates. This is the only bias of the toy.
- The **matcher** is a logistic regression on the two scores and the distance, trained on the past decisions. It does not read the region.

We apply the techniques of Table 21: reweighing (pre-processing), the Prejudice Remover with trade-off $\eta = 20$ (in-processing), and equalized odds post-processing, once on the hire or reject decisions and once on the score of the matcher. Every decision is evaluated on the test set against the truth (Table 23 and Figure 8).

Table 23. The toy case on the test set, against the truth. Accuracy, selection rates and error rates in %, SPD in percentage points. SPD and DI are those of the South against the North.

method	accuracy	rate North	rate South	SPD	DI	FNR North	FNR South	FPR North	FPR South
baseline	86.5	46.5	37.1	-9.4	0.80	15.2	29.5	8.2	2.6
reweighing	87.5	45.1	40.4	-4.7	0.90	16.4	24.0	6.6	3.7
Prejudice Remover	87.5	42.8	42.6	-0.2	0.99	19.3	20.8	4.9	4.8
equalized odds, decisions	82.5	39.9	40.0	0.1	1.00	27.4	28.1	7.2	7.1
equalized odds, score	88.2	49.7	49.2	-0.5	0.99	11.7	13.8	11.1	11.0

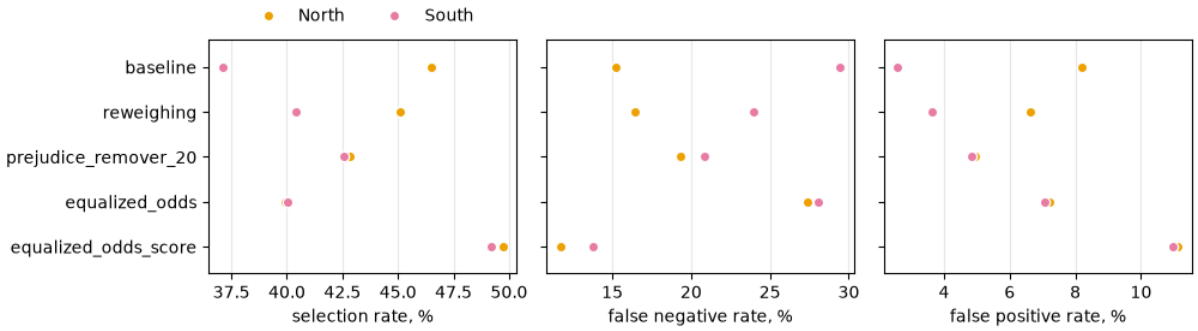


Figure 8. Selection rate and error rates of the two regions for every method of the toy case.

- The baseline never reads the region, but it learns the bias through the distance, so fairness through unawareness fails: the South is hired less often, its DI is at the limit of the four-fifths rule, and the matcher misses many more qualified southern candidates than northern ones.
- Reweighing halves the gaps in the selection rate and in both error rates. Part of the bias remains: the weights balance the labels between the regions only on average, and the wrongly rejected southern candidates keep their wrong label, so the matcher still learns from them.
- The Prejudice Remover closes all three gaps and is more accurate than the baseline. It closes them from both sides: in the North it misses more qualified candidates and hires fewer unqualified ones, in the South the opposite.
- Equalized odds on the decisions alone has the lowest accuracy. A random flip adds no information, so it mostly lowers the North, as in Section 7.1. On the score it has the highest accuracy: it lowers the false negative rate of both regions by hiring more in both, at the price of a higher false positive rate.

8 Discussion

8.1 Answers to the research questions

RQ1. The parser's errors reach the selection: correcting them changes the score of about one pair in four and the outcome of 8.3% of the pairs. They do not move places between groups: independence holds for gender, macro-region and their intersection, with Adecco's score, with O and with C. For the macro-region this is expected, because almost every list has one region only. There is no gender gap in the error rates. The South is penalized in a different way: the decisions on southern candidates are wrong more often in both directions, and confirmed less often. With both outcomes, the higher FNR and the lower PPV of the South remain. For women of the South we find gaps of the same size as the South's, with one outcome only: we have no evidence of an intersectional effect beyond the regional one.

RQ2. On the real system the mitigation techniques have little to act on: nothing is trained, the selection is a quota per list, and there is no ground truth. The one technique we could test, the post-processing of the decisions, closes the gap by leveling the North down: the cost is paid by northern candidates and no southern candidate gains. On the toy case all three techniques reduce the gap, and the cost depends on the information the technique can use.

RQ3. A parser error changes a selection when it moves a pair across the cut of its list. This depends on three things:

- The field of the error. Errors in the job titles matter most: `Experience` weighs the most, and job titles are the only field of the parser that enters the real profile.
- The other candidates. An error also changes the selection of the other candidates of the list: 44% of the pairs that change outcome keep the same score, and change only because a competitor's CV was corrected.
- The list. Where the score cannot separate the candidates, the pairs sit close to the cut and a small change is enough.

The South is hit more often because of its lists. The correction changes the score of southern and northern pairs equally often (Table 10), but southern lists more often require no skill that separates the candidates (Table 19). Being close to the cut explains about half of the South's gap (Section 6.2), and random noise in place of the parser's errors flips more southern decisions too (Section 6.4).

On robustness, P agrees with Adecco's decisions on 97% of the pairs, which validates the reconstruction, and the South's higher FNR and lower PPV hold with the top 3.

8.2 Limits

- **C is not the truth.** It holds what our extractors recovered. The errors we measure are those of O relative to C.
- **The outcome is ours.** "Top half of the list" is our definition, not a decision taken by Adecco. A job's list is not the real competition: a candidate is in it only if the job is among those recommended to them.
- **Small groups.** The Center and the South have a few hundred candidates each, so their intervals are wide.

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